

Distinct Nucleotide Binding/Hydrolysis Properties and Molar Ratio of MutS α and MutS β Determine Their Differential Mismatch Binding Activities*

Received for publication, February 9, 2009, and in revised form, February 18, 2009 Published, JBC Papers in Press, February 19, 2009, DOI 10.1074/jbc.M900908200

Lei Tian, Liya Gu, and Guo-Min Li¹

From the Graduate Center for Toxicology and Markey Cancer Center, University of Kentucky College of Medicine, Lexington, Kentucky 40536

MutS α (MSH2/MSH6) and MutS β (MSH2/MSH3) are eukaryotic mismatch recognition proteins that preferentially process base-base and small insertion/deletion (ID) mispairs, respectively, despite the fact that cells contain a MutS α :MutS β ratio of 10:1. To explore the mechanism underlying the differential mismatch recognition by these two proteins, purified human MutS α and MutS β were analyzed individually and competitively for their abilities to interact with a T-G and an ID substrate. We show that MutS α has K_D values of 26.5 and 38.2 nM for the G-T and ID substrates, respectively, and that MutS β has K_D values of 76.5 and 23.5 nM for G-T and ID, respectively. Consistent with these results, competitive binding assays revealed the following relative binding affinities: MutS β -ID > MutS α -T-G > MutS α -ID $\gg$ MutS β -T-G. Interestingly, binding of MutS β to ID heteroduplexes is greatly stimulated when the MutS α :MutS β ratio is ≥ 10 . Distinct ATP/ADP binding and ATPase activities of MutS α and MutS β were also observed. In the absence of DNA, ADP binding and ATPase activities of MutS β are significantly higher than those of MutS α . However, interaction with DNA significantly stimulates the MutS α ATPase activity and reduces the MutS β ATPase activity, the consequence being that both proteins exhibit the same level of hydrolytic activity. We conclude that the preferential processing of base-base and ID heteroduplexes by MutS α and MutS β is determined by their significant differences in ATPase activity, ADP binding activity, and high cellular MutS α :MutS β ratio.

DNA mismatch repair (MMR)² plays an important role in maintaining genome stability by primarily correcting both base-base mismatches and insertion/deletion (ID) mispairs generated during DNA replication (1). The MMR system is highly conserved as both prokaryotes and eukaryotes use a similar group of protein factors and a similar repair mechanism for correcting mispairs. MMR reactions in *Escherichia coli* and human cells have been reconstituted with purified proteins (2–4), which essentially involve mismatch recognition by MutS family proteins, removal of mispaired base by nucleases in a

manner dependent on MutL family proteins and several other protein factors, and DNA repair synthesis by replicative DNA polymerases in concert with factors involved in DNA replication (1).

Mismatch recognition is a critical step of MMR. In *E. coli*, recognition of both base-base and ID mismatches is conducted by the MutS protein. However, at least two mismatch recognition proteins, MutS α (the MSH2/MSH6 heterodimer) and MutS β (the MSH2/MSH3 heterodimer), have been identified in eukaryotic cells, and each of them is a heterodimeric complex. Both genetic and biochemical studies suggest that MutS α and MutS β have partially overlapping functions, with MutS α targeting base-base mismatches and 1–2-nucleotide (nt) ID mispairs and MutS β targeting ≥ 2 -nt but ≤ 16 -nt ID heteroduplexes (5–8). A recent genetic study in yeast suggests that MutS β may also play some role in the repair of base-base mismatches (9). Interestingly, cells make ~ 10 -fold more MutS α than MutS β , and overexpression of MSH3 results in a strong mutator phenotype (10, 11), presumably because the excess MSH3 saturates the pool of MSH2, essentially depleting MutS α in cells. It is unclear how MutS α and MutS β , at a 10:1 ratio, partition in cells to specifically process their favored substrates.

In addition to the mismatch binding activity, all MutS proteins, from *E. coli* to humans, contain an ATPase activity and ATP and ADP binding activities (12–14). Both the nucleotide binding and ATPase activities of the MutS family proteins are essential for MMR (15–19), but how these activities work in MMR is not fully understood. MutS family proteins can simultaneously bind ATP and ADP (20, 21) and undergo ADP $\rightarrow$ ATP exchange (8, 22–25) to induce MutS conformational changes, signaling downstream repair events. Mazur *et al.* (26) have recently demonstrated that two subunits of yeast MutS α exhibit differential nucleotide binding ability: the MSH6 subunit has a higher affinity for ATP binding than the MSH2 subunit, but the MSH2 subunit exhibits a higher affinity for ADP binding than the MSH6 subunit. ATP hydrolysis by MutS proteins is thought to promote translocation of these proteins along DNA helices (27) or to verify MutS mismatch binding and authorize the eventual repair reaction (18). Although much of the work concerning ATP/ADP binding and hydrolysis is conducted with bacterial MutS and eukaryotic MutS α , it is not known whether or not these activities in MutS β are different from those of MutS α and whether or not

* This work was supported, in whole or in part, by National Institutes of Health Grants GM072756 and CA115942 (to G.-M. L.) and CA104333 (to L. G.).

¹ James-Gardner Chair in Cancer Research. To whom correspondence should be addressed. E-mail: gmli@uky.edu.

² The abbreviations used are: MMR, mismatch repair; ID, insertion/deletion mismatch; nt, nucleotide.

Distinct Biochemical Activities between MutS α and MutS β

they contribute to specific recognition of ID heteroduplexes by MutS β .

To address these issues, purified human MutS α and MutS β were analyzed individually and competitively for their ability in base-base and ID mismatch recognition, ATP/ADP binding, and ATP hydrolysis. We identified some hitherto unknown properties of these two mismatch recognition proteins and their striking differences in nucleotide binding and ATPase activities. The possible involvement of these novel properties and activities in differential mismatch recognition by MutS α and MutS β is discussed.

EXPERIMENTAL PROCEDURES

Expression and Purification of Recombinant MMR Proteins—Human MutS α and MutS β were expressed in insect cells using the baculovirus system. Baculovirus stocks for the human *MSH2*, *MSH6*, and *MSH3* genes were generous gifts of Josef Jiricny (University of Zurich). MutS α and MutS β were overexpressed and purified essentially as described previously (4, 17). The recombinant proteins were purified to near homogeneity. Protein concentrations were determined by the Bio-Rad protein assay kit.

Gel Shift Analysis—Gel shift assays were performed in 20- μ l reactions containing 10 mM HEPES-KOH (pH 7.5), 110 mM KCl, 1 mM EDTA, 1 mM dithiothreitol, 32 P-labeled oligonucleotide heteroduplexes, and MutS α and/or MutS β in the presence of a 10-fold excess amount of unlabeled oligonucleotide homoduplex. The reactions were incubated on ice for 20 min, followed by the addition of 5 μ l of 50% (w/v) sucrose. Samples were loaded on and separated by electrophoresis through a 6% nondenaturing polyacrylamide gel in buffer containing 50 mM Tris borate (pH 7.5) and 1 mM EDTA. The buffer was recirculated during electrophoresis (28). The gel was dried and analyzed by a Storm PhosphorImager (GE Healthcare).

Nucleotide UV Cross-linking and ATPase Analyses—The nucleotide cross-linking assays were performed essentially as described (26). Reactions were assembled and incubated on ice in nucleotide binding buffer containing 50 mM Tris-HCl (pH 8.0), 110 mM NaCl, 2 mM dithiothreitol, 100 mg/ml bovine serum albumin, 0.5 mM EDTA, and 5% glycerol in the presence or absence of 5 mM MgCl₂. Where specified, DNA heteroduplex or homoduplex was added 10 min prior to addition of nucleotide. MutS α or MutS β was mixed with [α - 32 P]ATP, [α - 32 P]ADP, or [γ - 32 P]ATP and incubated for 10 min. Samples were then subjected to 10 min of UV cross-linking (Stratalinker), followed immediately by fractionation by 8% SDS-PAGE. Radiolabeled bands were quantified using a PhosphorImager. [α - 32 P]ADP was generated by incubating [α - 32 P]ATP with hexokinase and purified as described (26). ATPase activity of MutS α and MutS β was assayed in 20- μ l reactions containing 50 mM Tris-HCl (pH 8.0), 110 mM NaCl, 0.5 mM EDTA, 5 mM MgCl₂, [γ - 32 P]ATP, and the indicated amount of proteins and DNA substrates. After incubation at 37 °C for 10 min, the reactions were terminated and fractionated through a 20% denaturing polyacrylamide gel. 32 P-containing species were detected and quantified by PhosphorImager.

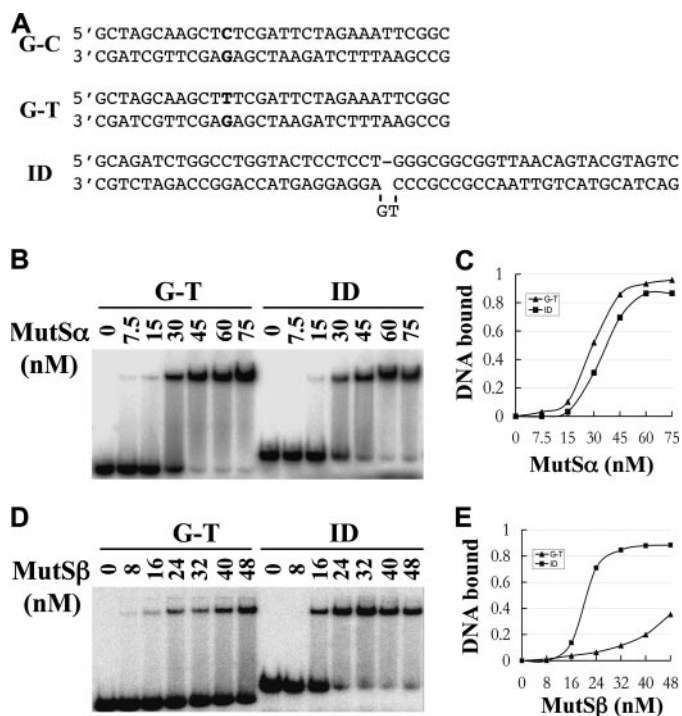


FIGURE 1. Interactions of MutS α and MutS β with G-T and ID heteroduplexes. A, oligonucleotide duplexes used in this study. B and C, interactions of MutS α with G-T and ID substrates. D and E, interactions of MutS β with G-T and ID substrates. Protein-DNA interactions were performed in 20- μ l reactions containing the indicated concentration of MutS proteins and 5 nM oligonucleotide duplex. The reactions were incubated on ice for 20 min, and the products were analyzed by gel shift assays. Representative gel shift analyses for MutS α and MutS β are shown in B and D, respectively, and the relative binding activities of individual reactions determined using the average binding value of two independent experiments are plotted in C and E.

RESULTS

Binding of a G-T and a 2-nt ID DNA Substrate by MutS α and MutS β —To examine the molecular basis by which MutS α and MutS β play differential roles in the repair of base-base mismatches and ID mispairs, purified recombinant human MutS α and MutS β were examined for their ability to interact with a 31-mer oligonucleotide duplex containing a G-T mismatch and a 50-mer duplex containing a 2-nt ID mispair, which are referred to as G-T and ID, respectively (Fig. 1A). In initial studies, the binding constants were determined for each protein/DNA substrate pair. Binding reactions were carried out in reactions with a constant DNA substrate concentration (5 nM) and a variable concentration of MutS α (7.5–75 nM) or MutS β (16–48 nM). Reaction products were visualized by the gel shift method and quantified using a PhosphorImager. Representative gel shift assays are shown in Fig. 1 (B and D). The fraction of bound and unbound DNA substrate was determined, and the values were plotted (Fig. 1, C and E).

Steady-state binding analysis showed that the K_D values of MutS α for the G-T and ID substrates were 26.5 and 38.2 nM, respectively. For MutS β , the K_D values for G-T and ID DNA substrates were 76.5 and 23.5 nM, respectively. Thus, MutS α has an ~1.5-fold higher affinity for the base-base mismatched substrate than for the ID substrate, whereas MutS β has an ~3-fold higher affinity for the ID substrate than for the base-base mismatched substrate. To confirm this result, each protein

was co-incubated with the G-T and ID DNA substrates in competition with each other by gel shift analysis (Fig. 2). Binding reactions were carried out in reactions with 6 nM MutS α and different ratios of the two DNA substrates (*i.e.* 5 nM G-T and 0–40 nM ID substrate or 5 nM ID substrate and 0–40 nM G-T DNA substrate). As shown in Fig. 2A, although the increase in the ID:G-T ratio is associated with the increased amount of free G-T substrate (lanes 5–8), only <50% of the free G-T probe is seen in the presence of an 8-fold excess amount of the ID substrate (compare the amount of free G-T substrate between lanes 1 and 8), indicating that at least 50% of the G-T substrate remains bound under this condition. When the substrate ratio was reversed (Fig. 2A, lanes 9–11), unbound ID substrate was almost kept at the same level as the input (compare the amount of free ID substrate in lanes 9–11 with that in lane 2). These results are consistent with the K_D values of MutS α for these two substrates described above and support the notion that MutS α preferentially binds to base-base mismatches.

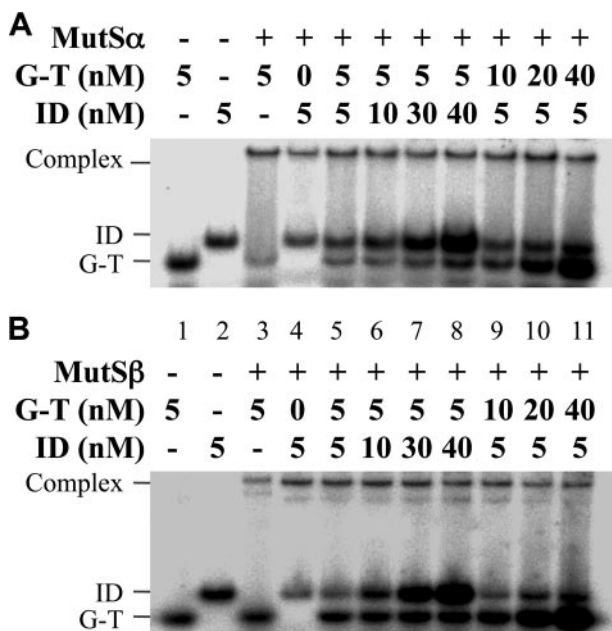


FIGURE 2. Competitive binding of MutS α or MutS β to T-G and ID heteroduplexes. Gel shift analyses were performed as described under "Experimental Procedures" using the indicated proteins and heteroduplexes. A, competitive binding of T-G and ID to MutS α ; B, competitive binding of T-G and ID to MutS β .

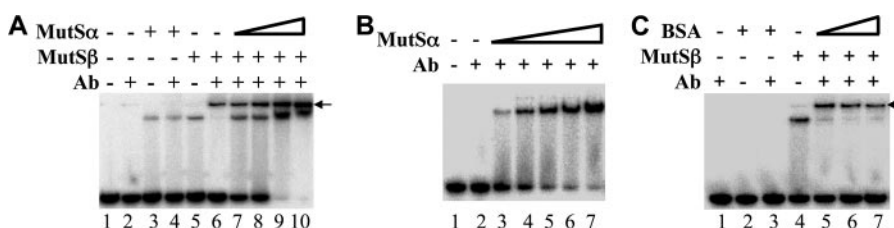


FIGURE 3. High MutS α :MutS β ratios stimulate MutS β -ID interaction. Unless otherwise specified, gel shift assays were performed (see "Experimental Procedures") using 20 nM MutS α and 16 nM MutS β . An antibody (Ab; 400 ng) against the MSH3 subunit of MutS β was used, as indicated, to supershift the MutS β -ID complex. A, stimulation of the MutS β -ID interaction (arrow) by high concentrations of MutS α . The MutS α concentrations used were 40, 80, 160, and 240 nM in lanes 7–10, respectively. B, the MSH3 antibody does not supershift the MutS α -ID complex. The MutS α concentrations used were 20, 40, 60, 80, and 100 nM in lanes 3–7, respectively. C, bovine serum albumin (BSA) at high concentrations does not stimulate the MutS β -ID interaction. The bovine serum albumin concentrations used were 80, 160, and 320 nM in lanes 5–7, respectively.

In competitive binding reactions with MutS β (Fig. 2B), the ID DNA substrate is bound preferentially in the examined conditions. When the two DNA substrates are equal in concentration (Fig. 2B, lane 5), the amount of unbound ID substrate is about the same (if not less) as that in the reaction containing only the ID substrate (lane 4); almost all G-T substrates exist in unbound form when excess amounts of ID are present (lanes 6–8). In contrast, a molar excess of the G-T substrate only slightly reduces the fraction of the ID DNA substrate bound by MutS β (Fig. 2B, lanes 9–11). Comparing the corresponding reactions in Fig. 2 (A and B) also draws a clear conclusion that the preferred substrate for MutS α or MutS β is the G-T or ID heteroduplex, respectively. These results are consistent with the K_D values for the enzyme/DNA substrate pairs noted above.

High MutS α :MutS β Ratios Stimulate MutS β Binding to ID Substrates—Cells express both MutS α and MutS β , and the two proteins may compete for binding to the same DNA heteroduplexes, especially those that are well recognized by both proteins. To simulate the situation *in vitro*, MutS α and MutS β were co-incubated with a 2-nt ID substrate, and the reaction products were analyzed by gel shift analysis. Protein-DNA complexes with MutS α and MutS β were distinguished by their ability to be "supershifted" by an anti-MSH3 antibody, which specifically supershifts the MutS β -DNA complex but not the DNA substrate and/or the MutS α -DNA complex (Fig. 3 and data not shown). Surprisingly, increasing amounts of MutS α stimulate binding of MutS β to the ID DNA substrate. When a reaction contained a MutS α :MutS β ratio ≥ 10 (Fig. 3A, lanes 9 and 10), >3-fold MutS β -DNA complex (see arrow) supershifted by the anti-MSH3 antibody was observed (compare lanes 9 and 10 with lane 6). No supershifted products were detected in the same reactions without MutS β (Fig. 3B), consistent with the fact that the antibody is highly specific to MutS β . The enhanced interaction between MutS β and the ID heteroduplex appears to be specifically mediated by MutS α because the addition of bovine serum albumin, regardless of the amount of protein used, did not promote binding of MutS β to the ID substrate (Fig. 3C). Therefore, these observations suggest that a MutS α :MutS β ratio ≥ 10 is necessary to stimulate MutS β affinity for its preferred DNA substrates.

MutS α and MutS β Possess Distinct Nucleotide Binding Activities—MutS protein family members share a conserved ATP/ADP-binding site and ATPase activity. Previous studies have shown that binding to ATP/ADP and hydrolysis of ATP by MutS or MutS α play a crucial role in MMR, including verifying mismatch recognition and authorizing the repair (18) or signaling protein translocation along the DNA molecule to initiate mismatch excision (23, 27). ATP hydrolysis by the MutS family ATPase requires two important cofactors: DNA and Mg²⁺ (21, 24, 26, 29). To explore whether MutS α and MutS β possess differential ATP/ADP binding and hydrolysis activities, which may contribute to their distinct mis-

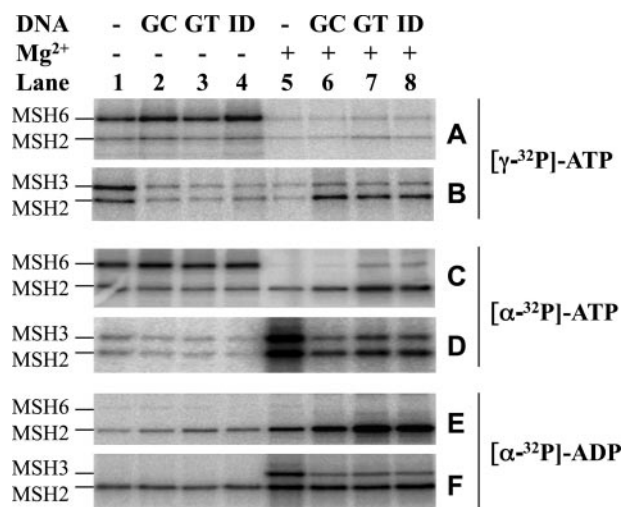


FIGURE 4. **Binding of MutS α and MutS β to ATP or ADP.** MutS α (A, C, and E) or MutS β (B, D, and F) was incubated with [γ - 32 P]ATP, [α - 32 P]ATP, or [α - 32 P]ADP, as indicated, in the presence or absence of the indicated DNA duplexes and 5 mM MgCl $_2$, followed by UV cross-linking and SDS-PAGE as described under "Experimental Procedures." The 32 P-cross-linked subunits were detected by a Storm PhosphorImager.

match recognitions, purified MutS α and MutS β were incubated in the presence of [γ - 32 P]ATP with or without DNA substrates. Bound ATP was immobilized by UV cross-linking, and reaction products were resolved by SDS-PAGE and visualized by a PhosphorImager. Under these conditions, the MSH6 subunit of MutS α is cross-linked much more efficiently to ATP than the MSH2 subunit of the protein (Fig. 4A, lanes 1–4), consistent with previous observations for yeast MutS α (26). In contrast, both the MSH2 and MSH3 subunits of MutS β are cross-linked to ATP with similar efficiency (Fig. 4B, lanes 1–4). Interestingly, whereas DNA duplexes, regardless of homoduplex (G-C) or heteroduplex (G-T or ID), have little effect on ATP binding to MutS α (Fig. 4A, lanes 1–4), they significantly reduce the MutS β -ATP interaction (compare lane 1 with lanes 2–4 in Fig. 4B). When the reactions were performed in the presence of Mg $^{2+}$, which supports ATP hydrolysis, little 32 P-labeled MutS α was detected in reactions containing [γ - 32 P]ATP (Fig. 4A, lanes 5–8), consistent with the fact that the 32 P-labeled phosphate (at the γ -position) is hydrolyzed by MutS α ATPase activity (26). However, under the same conditions, enhanced cross-links were observed in the MSH2 subunit of MutS β in the presence of DNA (Fig. 4B, lanes 6–8). This result suggests that MutS β , when interacting with DNA duplexes, has adapted a conformation in favor of ATP binding but not hydrolysis, and this seems to apply only to MSH2 but not MSH3 (Fig. 4B, lanes 6–8).

Similar cross-linking experiments were performed by substituting [γ - 32 P]ATP with [α - 32 P]ATP (Fig. 4, C and D). As expected, in the absence of Mg $^{2+}$ (*i.e.* no ATP hydrolysis), the amount of ATP cross-links to individual subunits of MutS α or MutS β is essentially the same as observed in reactions with [γ - 32 P]ATP (Fig. 4, compare lanes 1–4 in A and C for MutS α and in B and D for MutS β). Under conditions that support ATP hydrolysis (*i.e.* in the presence of Mg $^{2+}$), the MSH2 subunit but not the MSH6 subunit of MutS α was preferentially labeled (Fig. 4C, lanes 5–8), consistent with the observation with yeast

MutS α (26). In the case of MutS β , both subunits were well labeled, with a better cross-link for MSH2 (Fig. 4D, lanes 5–8). Apparently, DNA plays an inhibitory role in MutS β cross-linking with [α - 32 P]ATP, as judged by the fact that much intense labeling was detected for both MSH2 and MSH3 in the absence of DNA substrates (Fig. 4D, lane 5). Because Mg $^{2+}$ stimulates ATP hydrolysis and because DNA substrates selectively block Mg $^{2+}$ -provoked ATP hydrolysis by MutS β (Fig. 4B, lanes 6–8), the 32 P-labeled proteins in Fig. 4D could result from cross-linking to [α - 32 P]ATP (without hydrolysis), [α - 32 P]ADP (with hydrolysis), or both.

To distinguish these possibilities, cross-linking experiments were conducted in the presence of [α - 32 P]ADP. As shown in Fig. 4E, only the MSH2 subunit of MutS α interacts with ADP. Interestingly, this interaction is greatly enhanced in the presence of Mg $^{2+}$ (compare lanes 5–8 with lanes 1–4, respectively, in Fig. 4E), and the enhancement is more pronounced in reactions containing heteroduplexes (lanes 7 and 8). The cross-linking experiments performed with MutS β reveal that in the absence of Mg $^{2+}$, the protein behaves similarly to MutS α , *i.e.* only the MSH2 subunit cross-links to ADP (Fig. 4F, lanes 1–4); however, addition of Mg $^{2+}$ to the reaction not only stimulates the MSH2-ADP interaction but also promotes the MSH3 subunit to interact with the nucleotide (lanes 5–8). ADP appears to bind equally well to MSH2 and MSH3 in the presence of Mg $^{2+}$ and the absence of DNA (Fig. 4F, lane 5); DNA greatly reduces the affinity of MSH3 but not that of MSH2 for ADP (lanes 6–8). Similar to the interaction between MutS α and ADP, there appeared to be a little more ADP binding to the MSH2 subunit of MutS β in the reaction containing the ID substrate (Fig. 4F, lane 8). These results may explain why Mg $^{2+}$ is required for mismatch binding by MutS (29) and why MutS proteins in their ADP-bound form possess a higher affinity for heteroduplexes (18, 30). Comparing data in Fig. 4 (D and F), it appears that the cross-links in D (lanes 5–8) contain components of both ADP and ATP. These observations suggest that DNA stimulates the ATPase activity of MutS α , but it slightly inhibits the ATPase activity of MutS β .

MutS α and MutS β Possess Distinct ATPase Activities—The ATPase activity of MutS proteins is essential for their functions in MMR (17, 19). It has also been shown that the ATPase activity of MutS α could be stimulated by homo- or heteroduplex DNA although to different extents (20, 31, 32). To determine whether there is any difference in ATPase activity between MutS α and MutS β , which may contribute to their preferential mismatch recognition, the purified human MutS heterodimers were assayed for their ability to hydrolyze [γ - 32 P]ATP in the presence or absence of DNA substrates. The 32 P-containing species, *i.e.* the unreacted [γ - 32 P]ATP and the hydrolyzed [γ - 32 P]phosphate, were detected after gel electrophoresis (Fig. 5, A and B). The results indicate that in the absence of DNA substrates, MutS β exhibited a much more active ATPase activity than MutS α at all concentrations and time points tested (Fig. 5, C and D). However, DNA substrates, regardless of a homoduplex and a heteroduplex, significantly stimulated the ATPase activity of MutS α (Fig. 5D; also compare lane 7 with lanes 8–10 in Fig. 5A), consistent with previous observations. Surprisingly, DNA substrates were found to inhibit MutS β ATPase activity

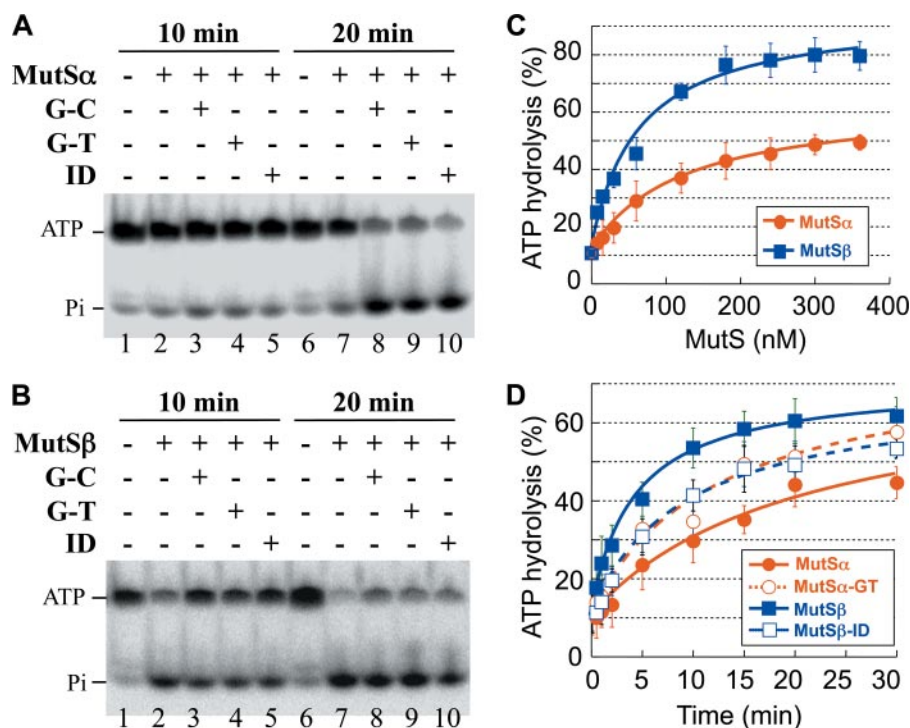


FIGURE 5. ATPase analysis of MutS α and MutS β . Unless otherwise specified, ATPase activity of MutS α or MutS β was assayed in reactions containing 50 nM proteins, [γ - 32 P]ATP, and 5 mM MgCl $_2$ in the presence or absence of the indicated DNA substrates. The reactions were incubated at 37 °C for the indicated times, followed by electrophoresis as described under "Experimental Procedures." 32 P-labeled species were detected and quantified by a PhosphorImager. A and B, representative ATPase assays for MutS α and MutS β , respectively; C, titration of ATPase activity of MutS α and MutS β ; D, ATPase activity in a time course. Pi, [32 P]phosphate.

by ~20% (Fig. 5D; also compare lane 7 with lanes 8–10 in Fig. 5B). These results differ somewhat from those of Fishel and co-workers (8) who reported stimulation of MutS β ATPase activity by ID substrates. Although the exact reason for this discrepancy is unknown, we did notice that a His-tagged MutS β and a nontagged MutS β were used in the previous study and this study, respectively, which may have an impact on MutS β ATPase activity. Interestingly, despite the stimulation of MutS α activity and the reduction of MutS β activity by DNA substrates, both proteins exhibited the same level of ATPase activity upon their interactions with DNA (see Fig. 5D), suggesting that the DNA-associated ATPase activity of MutS proteins is not related to mismatch binding specificity but to the downstream signaling of MMR.

DISCUSSION

This study investigates molecular mechanisms by which MutS α and MutS β preferentially process base-base mismatches and ID mispairs using purified human MMR proteins. Some interesting observations have been made in this study, including enhancement of the MutS β -ID interaction by excess amount of MutS α and significant differences between MutS α and MutS β in DNA substrate recognition, ATP/ADP binding, and ATP hydrolysis. These differences may influence the functional roles of these two proteins in MMR *in vivo*.

One of the puzzling phenomena in MMR is that MutS β dominantly binds and directs repair of small ID mispairs, despite the fact that the amount of MutS β in human cells is only one-tenth the amount of MutS α (10) and that MutS α also recognizes ID

heteroduplexes (reviewed in Ref. 1). Our work presented here provides some significant insights into this question. First, our steady-state *in vitro* DNA binding studies reveal that MutS α and MutS β display distinct specificities for base-base and ID heteroduplex binding and have the following hierarchy of binding affinities: MutS β -ID > MutS α -G-T > MutS α -ID $\gg$ MutS β -G-T (K_D values were 23.5, 26.5, 38.2, and 76.5 nM, respectively). Second, we surprisingly find that MutS α at a high concentration does not inhibit but stimulates the binding activity of MutS β for ID heteroduplexes (Fig. 3). This finding explains why cells have to maintain a 10:1 MutS α :MutS β ratio and why MutS β at a low concentration is capable of efficiently processing ID heteroduplexes. Because MSH2 is shared between MutS α (MSH2/MSH6) and MutS β (MSH2/MSH3), the distinct MSH6 and MSH3 subunits compete against each other for MSH2 *in vivo*. Previous studies show that overexpression of MSH3

greatly reduces the MutS α :MutS β ratio, leading to a mutator phenotype (10, 11). This is apparently because base-base mismatches, which are poor substrates of MutS β (Fig. 1) (4), are left unrepaired under conditions of insufficient MutS α . Thus, the high ratio of MutS α to MutS β appears to be a mechanism ensuring efficient repair of both base-base and ID heteroduplexes, *i.e.* a high level of MutS α not only guarantees the efficient processing of base-base mismatches but also promotes the efficient repair of ID mispairs by stimulating MutS β activity.

However, the molecular basis as to how MutS α stimulates the MutS β affinity for ID heteroduplexes is unclear. Because multiple molecules of MutS proteins are required for processing a single mismatch (4), one possibility is that binding of MutS α to homoduplex DNA regions (*i.e.* unlabeled noncompetitive DNA in the case of the gel shift reactions) allows MutS β to focus on ID binding, resulting in a dramatic increase in the local concentration of MutS β for an efficient repair. It is also possible that MutS α and MutS β may physically interact with each other, and abundant MutS α proteins can facilitate the MutS β -ID heteroduplex interaction by initially localizing the ID mispairs and passing them to MutS β for a specific and efficient repair of the ID heteroduplexes (4). Further studies are required to define the molecular mechanism by which a high MutS α concentration enhances the MutS β -ID interaction.

Another important observation of this study is that the ATP/ADP binding and ATP hydrolysis characteristics of MutS α and MutS β are significantly different. Under the experimental conditions (pH 7.5; 5 mM Mg $^{2+}$ and 110 mM NaCl), MutS β pos-

sesses a higher ADP binding activity than MutS α regardless of the presence of DNA, and this characteristic is particularly reflected on the MSH3 subunit (see Fig. 4, lanes 5–8, compare C with D and E with F). Interestingly, the ADP binding activity of MutS α and MutS β appears to be correlated with their ATPase activity, as MutS β has a more active ATPase activity than MutS α in the absence of DNA (Fig. 5C). This correlation suggests that prior to interacting with DNA, MutS β has a high tendency to exist in an ADP-bound status, a form of MutS proteins in favor of heteroduplex binding (30). This explains why the MutS β -ID interaction is stronger than the MutS α -ID interaction. The differential biochemical activities of MutS α and MutS β may be determined by their ternary structures. The MutS α -DNA co-crystal structures revealed that the MSH2 nucleotide-binding site is surrounded by two well organized Walker P-loops, but the corresponding P-loops in MSH6 are partially disorganized (14), which may explain why MSH2 has a higher ADP binding activity than MSH6. Although the crystal structure of MutS β is not available at this time, we predict that the P-loops are better organized in MSH3 than in MSH6. We also find that binding to DNA reduces MutS β ATPase activity but stimulates MutS α ATPase activity; as a result, both proteins exhibit the same level of ATPase activity (Fig. 5D). These results suggest that whereas the ATPase of MutS proteins enhances their heteroduplex affinity by converting the ATP-bound proteins to their ADP-bound form before interacting with DNA, the hydrolytic activity, upon binding of MutS proteins to a heteroduplex, acts to signal downstream repair events, including recruiting other MMR proteins and authorizing the repair reaction (1, 18, 30).

In summary, this study demonstrates significant differences in the *in vitro* DNA binding, ATP/ADP binding, and ATP hydrolysis characteristics of human MutS α and MutS β . These differences, together with a high MutS α :MutS β ratio, are likely responsible for the preferential recognition and repair of base-base and ID mispairs by MutS α and MutS β , respectively. Although the environment in living cells (which includes many other components and conditions that are different from those in reconstituted assays with purified components) may modulate MMR in a more complex manner, the biochemical characteristics of MutS α and MutS β , as well as their relative concentrations in cells, appear to play an important role in determining their functions *in vivo*.

REFERENCES

- Li, G.-M. (2008) *Cell Res.* **18**, 85–98
- Lahue, R. S., Au, K. G., and Modrich, P. (1989) *Science* **245**, 160–164
- Constantin, N., Dzantiev, L., Kadyrov, F. A., and Modrich, P. (2005) *J. Biol. Chem.* **280**, 39752–39761

- Zhang, Y., Yuan, F., Presnell, S. R., Tian, K., Gao, Y., Tomkinson, A. E., Gu, L., and Li, G.-M. (2005) *Cell* **122**, 693–705
- Genschel, J., Littman, S. J., Drummond, J. T., and Modrich, P. (1998) *J. Biol. Chem.* **273**, 19895–19901
- McCulloch, S. D., Gu, L., and Li, G.-M. (2003) *J. Biol. Chem.* **278**, 3891–3896
- Palombo, F., Iaccarino, I., Nakajima, E., Ikejima, M., Shimada, T., and Jiricny, J. (1996) *Curr. Biol.* **6**, 1181–1184
- Wilson, T., Guerrette, S., and Fishel, R. (1999) *J. Biol. Chem.* **274**, 21659–21664
- Harrington, J. M., and Kolodner, R. D. (2007) *Mol. Cell. Biol.* **27**, 6546–6554
- Drummond, J. T., Genschel, J., Wolf, E., and Modrich, P. (1997) *Proc. Natl. Acad. Sci. U. S. A.* **94**, 10144–10149
- Marra, G., Iaccarino, I., Lettieri, T., Roscilli, G., Delmastro, P., and Jiricny, J. (1998) *Proc. Natl. Acad. Sci. U. S. A.* **95**, 8568–8573
- Lamers, M. H., Perrakis, A., Enzlin, J. H., Winterwerp, H. H., de Wind, N., and Sixma, T. K. (2000) *Nature* **407**, 711–717
- Obmolova, G., Ban, C., Hsieh, P., and Yang, W. (2000) *Nature* **407**, 703–710
- Warren, J. J., Pohlhaus, T. J., Changela, A., Iyer, R. R., Modrich, P. L., and Beese, L. S. (2007) *Mol. Cell* **26**, 579–592
- Alani, E., Sokolsky, T., Studamire, B., Miret, J. J., and Lahue, R. S. (1997) *Mol. Cell. Biol.* **17**, 2436–2447
- Dufner, P., Marra, G., Raschle, M., and Jiricny, J. (2000) *J. Biol. Chem.* **275**, 36550–36555
- Iaccarino, I., Marra, G., Palombo, F., and Jiricny, J. (1998) *EMBO J.* **17**, 2677–2686
- Junop, M. S., Obmolova, G., Rausch, K., Hsieh, P., and Yang, W. (2001) *Mol. Cell* **7**, 1–12
- Studamire, B., Quach, T., and Alani, E. (1998) *Mol. Cell. Biol.* **18**, 7590–7601
- Bjornson, K. P., Allen, D. J., and Modrich, P. (2000) *Biochemistry* **39**, 3176–3183
- Blackwell, L. J., Bjornson, K. P., and Modrich, P. (1998) *J. Biol. Chem.* **273**, 32049–32054
- Acharya, S., Foster, P. L., Brooks, P., and Fishel, R. (2003) *Mol. Cell* **12**, 233–246
- Gradia, S., Acharya, S., and Fishel, R. (1997) *Cell* **91**, 995–1005
- Gradia, S., Subramanian, D., Wilson, T., Acharya, S., Makhov, A., Griffith, J., and Fishel, R. (1999) *Mol. Cell* **3**, 255–261
- Mendillo, M. L., Mazur, D. J., and Kolodner, R. D. (2005) *J. Biol. Chem.* **280**, 22245–22257
- Mazur, D. J., Mendillo, M. L., and Kolodner, R. D. (2006) *Mol. Cell* **22**, 39–49
- Allen, D. J., Makhov, A., Grilley, M., Taylor, J., Thresher, R., Modrich, P., and Griffith, J. D. (1997) *EMBO J.* **16**, 4467–4476
- Gu, L., and Li, G.-M. (2006) *Methods Enzymol.* **408**, 303–317
- Biswas, I., and Hsieh, P. (1996) *J. Biol. Chem.* **271**, 5040–5048
- Fishel, R. (1999) *Nat. Med.* **5**, 1239–1241
- Bowers, J., Sokolsky, T., Quach, T., and Alani, E. (1999) *J. Biol. Chem.* **274**, 16115–16125
- Gradia, S., Acharya, S., and Fishel, R. (2000) *J. Biol. Chem.* **275**, 3922–3930